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# FORREST BICKER

Mathematics + Computer Science

in forrestbicker

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## EDUCATION

### Harvey Mudd College

*B.S. Mathematics and Computer Science*

- 4.0 Major GPA, ICPC, ML Research at {AMISTAD Lab, Backgammon Lab}

Claremont, CA

*Aug 2021 - May 2025*

### Hunter College High School

New York, NY

## EXPERIENCE



### Arrowstreet Capital

*Quantitative Researcher*

Boston, MA

*Jul 2025 - Sep 2026*

- One of 25 researchers on a long-horizon global systematic multi-asset team managing \$350B.



### DE Shaw & Co.

*Quant Intern*

New York, NY

*Jun - Aug 2024*

- Researched enhancements to portfolio optimization reward model.
- Used numerical methods to improve quadratic program solver efficiency.



### Amazon Web Services

*Software Engineer Intern*

Seattle, WA

*Jul - Sep 2022*

- Robustified AWS Kubernetes provisioning and recovery systems to orchestrate 10,000-node ultra-scale GPU clusters.

## RESEARCH & PROJECTS

### Backgammon Research Group

*Backgammon Research under Dr. Arthur Benjamin*

Claremont, CA

*Nov 2021 - May 2022*

- Designed quantitative strategies for tournament backgammon play under world champion Dr. Arthur Benjamin.
- Developed novel betting strategy that outperforms state-of-the-art methods and proved successful in championship play.

### AMISTAD Lab

*Machine Learning Research under Dr. George Montañez*

Claremont, CA

*May - Sep 2022*

- Proved theoretical bounds on the Bias-Expressivity tradeoff for continuous learning problems and fuzzy membership.
- Published “From Targets to Rewards: Continuous Target Sets in the Algorithmic Search Framework” in ICAART 2024

### Reinforcement Learning Research

*Self-Play RL Agents*

Claremont, CA

*Sep 2023 - May 2024*

- Built a reinforcement learning research system that placed top 3 globally out of 25,000 teams and 100,000 submitted algorithms in a machine learning competition.
- Notes: Convolutional transformer actor-critic architecture, auto-regressive action decoding, behavioral cloning on 1M expert games, Prioritized Fictitious Self Play league RL across 20 million episodes, parallelized PPO with concurrent rollouts across 72 workers and async GPU learner, shallow MCTS for efficient inference time compute.

## AWARDS

**1st Place Paper** at Citadel Global Data Open Championship (\$100,000 prize)

**1st Place** at IMC Prosperity Global Trading Challenge (\$25,000 prize, 1st of 13,500)

**Top 50 USA** at ICPC North American Championship (International Collegiate Programming Contest NAC-NAPC)

**1st Place (2x)** at West Coast Datathon (\$20,000 prize)

**1st Place** at National Security Agency (NSA) Cybersecurity Hacklytics Challenge

**3rd Place** Algorithm at Worldwide Correlation One Terminal (\$2,500 prize, 3rd of 25,000)

## SKILLS

🐍 Python, 🔥 Pytorch, 📦 NumPy, 📊 Pandas, 📈 Statsmodels, 🏗️ CVXPY